



INSTALLATION MANUAL

DUCTED SYSTEM R-32 Series

Air Conditioners

MODELS

(Ceiling concealed duct type)

FDYAN50/60/71/85/100/125/140/160AV1

FDYA71/85/100/125/140/160AV1

PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE INSTALLATION.
KEEP THIS MANUAL IN A HANDY PLACE FOR FUTURE REFERENCE.

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Installation Manual

FDYAN50/60/71/85/100/125/140/160AV1

FDYA71/85/100/125/140/160AV1



Read the precautions in this manual carefully before operating the unit.



This appliance is filled with R32.

SAFETY CONSIDERATIONS

- Keep this manual where the user can easily find it.
- The precautions described herein are classified as WARNING and CAUTION. They both contain important information regarding safety. Be sure to observe all precautions without fail.

 **WARNING**..... Failure to observe a warning may result in death, injury or damage to the equipment.

 **CAUTION**..... Failure to observe a caution may result in injury or damage to the equipment.

WARNING

- Ask your Dealer or qualified personnel to carry out installation work. Do not try to install the machine yourself. Improper installation may result in water leakage, electric shock or fire.
- Perform installation work in accordance with this installation manual. Improper installation may result in water leakage, electrical shock or fire.
- Be sure to use only the specified accessories and parts for installation work. Failure to use the specified parts may result in water leakage, electric shock, fire or the unit failing.
- Install the conditioner on a structure strong enough to withstand the weight of the unit. A structure of insufficient strength may result in the equipment falling and causing injuries.
- Carry out the specified installation work after taking into account strong winds, typhoons or earthquakes. Improper installation work may result in the equipment falling and causing accidents.
- Make sure that a separate power supply circuit is provided for this unit and that all electrical work carried out by qualified personnel according to local laws, regulations and this installation manual. An insufficient power supply capacity or improper electrical construction may lead to electric shocks or fire.
- Make sure that all wiring is secured, the specified wires are used, and no external forces act on the terminal connection or wires. Improper connections or installation may result in fire.
- When wiring the power supply and connecting the wiring between the indoor and outdoor units, position the wires so that the switch box cover can be securely fastened. Improper positioning of the switch box cover may result in electric shock, fire or the terminals overheating.
- If the refrigerant gas leaks during installation, ventilate the area immediately.
- After completing the installation work, check that the refrigerant gas does not leak. Fire may happen if leaked refrigerants is in contact with ignition source.
- Before obtaining access to the terminals and electrical parts, all supply circuits must be disconnected.
- Do not install the indoor unit or remote controller where they may be exposed to rain or moisture. Water or other fluids on the electrical components may result in fire or electric shock.
- Earth the air conditioner. Do not connect the earth wire to gas or water pipes, a lightning conductor or a telephone ground wire. Improper earthing may result in electric shock.
- Be sure to install an earth leakage breaker. Failure to install an earth leakage breaker may result in electric shock.
- Do not install the indoor unit or remote controller where flammable gases may leak, where there are carbon fibre or ignitable dust suspended in the air, or where volatile flammables such as thinner or gasoline are handled.
- Do not install the indoor unit or remote controller where the area is filled with steam or the ground is always wet. Insulation of the electric components may be damaged in such conditions and may result in electric shock.
- Be sure to only use refrigerant piping approved for use with R32 refrigerant. The use of unapproved piping may result in explosive rupture.

CAUTION

- Do not install the air conditioner for purpose other than for human comfort. Do not use the air conditioner for special applications such as the storage of foods, animals and plants, precision machines and art objects. Deterioration of quality may result.
- This model is not designed for free blow operation. Supply and Return air duct work must be installed. External duct work and air boxes must be adequately insulated to prevent condensation.
- It is recommended that a safety drain tray is installed. Ensure that it drains separately.
- While following the instructions in this installation manual, install drain piping in order to ensure proper drainage and insulation piping in order to prevent condensation. Improper drain piping may result in water leakage and property damage.
- Install the indoor and outdoor units, power cords and connecting wires at least 1 metre away from televisions or radios in order to prevent image interference or noise. (Depending on the radio waves, a distance of 1 metre may not be sufficient enough to eliminate noise).
- Do not install the air conditioner in the following locations:
 - (a) Where a mineral oil mist, oil spray vapour is produced, for example in a kitchen. Plastic parts may deteriorate.
 - (b) Where corrosive gas, such as sulfuric acid gas, is produced. Corroding copper pipes or soldered parts may result in refrigerant leakage.
 - (c) Near machinery emitting electromagnetic waves. Electromagnetic waves may disturb the operation of the control system and result in a malfunction of the equipment.
 - (d) Where salinity in the air is relatively high.
- Be sure to attach a return air filter (field supplied) inside the return air passage. Failure to install the filter will cause equipment failure.

BEFORE INSTALLATION

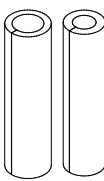
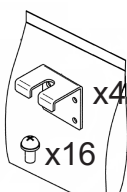
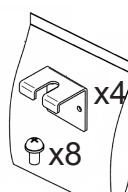
- The accessories needed for installation must be retained in your custody until the installation work is completed. Do not discard them!
- Decide upon a safe line of transport.
- Try to leave the unit inside its packaging whilst moving. Where unpacking is unavoidable use a sling of soft material or protective plates together with a rope when lifting to avoid damage or scratches to the unit.
- When opening the unit or moving it after opening be sure to lift it by holding onto the unit carefully without exerting any pressure on other parts, especially drain piping, refrigerant piping and other resin parts.
- Be sure to check the type of refrigerant to be used before doing any work. (Using an incorrect refrigerant will prevent normal operation of the unit.) This model uses R32.
- This product contains Refrigerant type: R32, which have GWP(Global Warning Potential) value : 675.
- R32 Classification as Flammable Refrigerant with Category 1 Flammable Gas (in WHS Regulations), and Division 2.1 Flammable Gas in Transport Code.
- Ventilate the area in the event of any refrigerant leak, Pressure test the system to 4.0MPa, and Evacuate the system to a minimum of 500microns, remove any potential source of ignition and have a fire extinguisher at hand.
- Refer to the installation manual of the outdoor unit in conjunction with this manual for items not described in this manual.

PRECAUTIONS

- Be sure to read this manual before installing the indoor unit.
- This unit is for indoor use only.
- This unit is suitable for installation in a household, commercial and light industrial environment.
- Do not install or operate the unit in rooms mentioned below.
 - Laden with mineral oil or filled with oil vapour or spray like in kitchens.
 - Where corrosive gas like sulfuric gas exists.
 - Where volatile flammable gas like thinner or petrol is used.
 - Where machines can generate electromagnetic waves.
 - Where the air contains high levels of salt or other corrosive materials.
 - Where voltage fluctuates greatly.
 - In vehicles or vessels.
- To avoid water leaks, etc., be sure to install a return air filter (field supplied) inside the return air passage.

ACCESSORIES

Check that the following accessories are included with your unit.

Name	Hose Clamp	Drain Hose	Installation Manual	Operation Manual	Pipe Insulation	Hanger Bracket Accessories Pack (140/160 only)	Hanger Bracket Accessories Pack (Others)
Quantity	1	1	1	1	2	1	1
Shape							

Note:

- Hanger Brackets Accessories Pack is only supplied for models FDYA(N)71-160AV1, consists of 4 hangers brackets and M6 assembly screws for the hanger brackets.

OPTIONAL ACCESSORIES

- A Daikin remote controller is not included in the packaging with the indoor unit. It needs to be purchased separately. Install the remote controller in an appropriate place.
If utilizing the temperature sensor in the remote controller BRC1E62/63 (see page 14) ensure that the controller is installed in a location where the correct air temperature is measured.
- Hanging brackets are required if the indoor unit is to be suspended. (50/60 Models).

NOTE TO THE INSTALLER

Be sure to instruct the customer how to properly operate the system (especially cleaning filters, operating the different functions and adjusting the temperature). It is recommended to ask the customer to carry out the operations themselves while you instruct them and refer them to the applicable place in the operation manual.

FOR THE FOLLOWING ITEMS, TAKE SPECIAL CARE DURING CONSTRUCTION AND CHECK AFTER INSTALLATION IS FINISHED.

(1) Items to be checked after completion of work.

Items to be checked	If not properly done, what is likely to occur	Check
Is the indoor unit fixed firmly?	The unit may drop, vibrate or make noise.	
Is the gas leak test finished?	It may result in insufficient cooling or heating.	
Does the unit require external insulation?	Condensate water may drip.	
Does drainage flow smoothly?	Condensate water may drip.	
Does the power supply voltage correspond to that shown on the name plate?	The unit may malfunction or the components burn out.	
Is the wiring and piping correct?	The unit may malfunction or the components burn out.	
Is the unit safely grounded?	Dangerous during electric leakage.	
Is the wiring size according to specification?	The unit may malfunction or the components burn out.	
Is something blocking the air outlet or inlet of either the indoor or outdoor units?	It may result in insufficient cooling or heating.	
Are refrigerant piping length and additional refrigerant charge noted down?	The refrigerant charge in the system is not clear.	
Is the indoor unit positioned so that it is not in contact with parts of the ceiling?	Vibration and noise can be transmitted into building.	
Is the airflow within specification?	Poor performance or malfunction.	
Is an air filter installed in the return air?	Blocked heat exchanger causing malfunction or compressor failure.	
Is an external drain tray installed?	Condensate water may drip.	

(2) Items to be checked at completion

- Did you show the customer how to operate the system while referring to the operation manual?
Detailed explanation about operation.
Review the safety consideration.
- Did you hand the operation manual over to the customer?

SELECTING INSTALLATION SITE

(1) Select an installation site where the following conditions are fulfilled and that meets your customer's approval.

- Where the indoor unit and remote controller are not exposed to rain or moisture.
- Where the casing of the unit does not contact parts of the ceiling.
- Where optimum air distribution can be ensured.
- Where nothing blocks the air passage.
- Where condensate can be properly drained.
- Where the structure is strong enough to safely bear the indoor unit weight.
- Where sufficient clearance for maintenance and service can be ensured. (see Fig. 1)
- Where there is no risk of flammable gas leakage.

- Where piping between indoor and outdoor units is possible within the allowable limit. (Refer to the installation manual for the outdoor unit.)

CAUTION

- Keep the indoor unit, outdoor unit, power supply wiring and transmission wiring at least 1 metre away from televisions and radios. This is to prevent image interference and noise in those electrical appliances. (Noise may be generated depending on the conditions under which the electrical wave is generated, even if 1 metre is kept.)
- (2) When using suspension bolts for installation, hanging brackets should be installed. Confirm that the ceiling is strong enough to support the weight of the indoor unit.

- Hanging brackets are sold as an optional accessory.
(For 50/60 models only)

- (3) If the installation requires the indoor unit to be exposed please ensure the unit is more than 2.5m above the floor so that children can not touch the unit.
- (4) A suitable air filter (field supplied) must be installed within the return air duct.

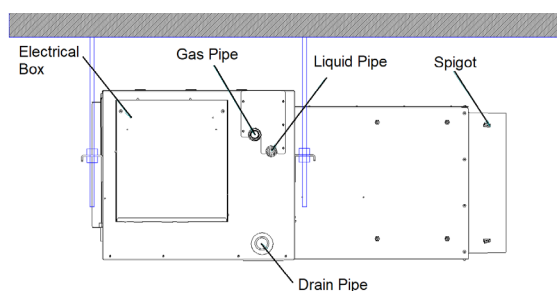


Fig. 2

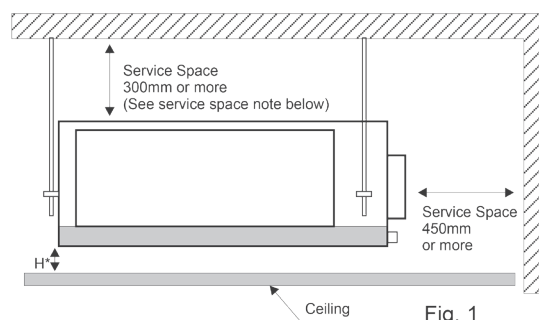


Fig. 1

Notes:

H^* (20-30 mm) is necessary to connect the drain pipe. Determine size H^* based on 1/100 of the required downward inclination indicated in "DRAIN PIPING WORK". Pg9

"Service Space" is the minimum recommended distance. In some "restricted space" installations it may be necessary to reduce this distance. Please note that this may necessitate additional service work should maintenance or repair be required.

NOTE: Additional thermal insulation material (thickness 10mm or more of glass wool or polyethylene foam) may be required if the conditions in the ceiling exceeds 30°C and RH 80% or outside air is introduced into the ceiling or the unit operates for 24 hours. It is also important that a safety drain tray be used. (see Fig. 11)

PREPARATION BEFORE INSTALLATION

To facilitate installation, the indoor unit can be easily separated into smaller parts to make it easier to move to the final position. See page 24 for details.

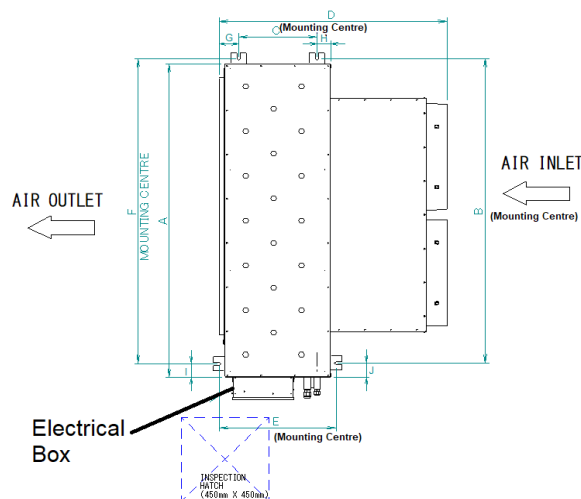


Fig. 3

Table 1

Model	A	B	C	D	E	F	G	H	I	J
FDYAN50/60AV1 FDYA(N)71/85/100AV1	1090	1072	354	900	460	1072	52	28	44	44
FDYAN125/140/160AV1, FDYA125AV1	1400	1365	347	935	460	1365	56	32	60	60
FDYA140/160AV1	1400	1365	340	980	502	1365	82	60	62	62

- (1) Refer to Table 1 and Fig. 3 for suspension bolt position and horizontal representation of the indoor unit.

- (2) Install the suspension bolt. (Fig. 3 & 4)
(Use M10 size rod for the suspension bolt)
Use an anchor suitable for both the material being anchored to and the weight of the unit.
- (3) Be sure to completely insulate all external duct work and fittings.
- (4) If using rigid duct work install a canvas (or similar) duct to the air inlet and air outlet so as not to transmit vibration from the unit to the duct.
Apply acoustic insulation to the inside of the duct.
- (5) Install suitable vibration isolator to the suspension bolts.

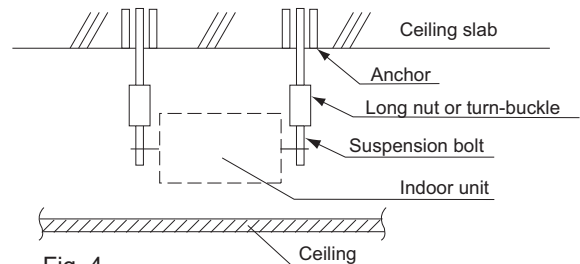


Fig. 4

NOTE: All above parts are field supplied

INDOOR UNIT INSTALLATION

- (1) For suspended installation, hanging brackets should be installed on the unit.
Attach the hanging bracket to the 8 mounting holes (16 mounting holes for models FDYA140/160AV1) located on the sides of the unit. (Fig. 3).
- (2) Carefully lift the unit so that the hanging brackets can locate on to the suspension bolts. (Fig. 5) Be sure to clamp the bracket securely by using a nut and washer (field supplied) from the upper and lower sides of the hanging bracket. Ensure that the suspension bolts are firmly secured. (No up or down movement) (Fig. 5)

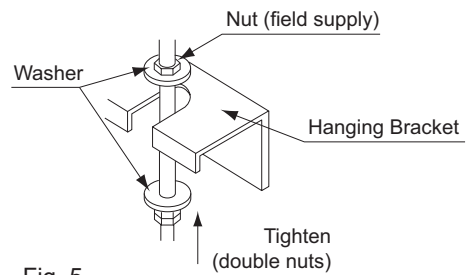


Fig. 5

⚠ CAUTION

- (3) Make sure the unit is level.
 - Level the unit with a water level or other suitable means. If the unit is not level, it could become the source of water leaks.
 - Check if the unit is level at all four corners with a water level or a water-filled vinyl tube as shown in the diagrams. (Fig. 6)
- (4) Tighten the upper nut.

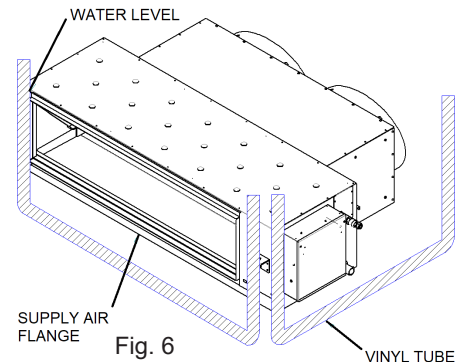


Fig. 6

REFRIGERANT PIPING WORK

ALL REFRIGERANT PIPING WORK MUST ONLY BE CARRIED OUT BY A QUALIFIED PERSON

**** For refrigerant piping of outdoor units, see the installation manual attached to the outdoor unit. ****

Fully insulate the liquid and gas piping separately. Do not leave any gaps otherwise a water leak may occur. (Seal any joints with glue or duct tape)

When operating in the heating mode, the temperature of the gas piping can reach up to approximately 120°C, so use insulation which has suitable temperature rating.

Also, in cases where the temperature and humidity surrounding the refrigerant piping can exceed 30°C or RH 80%, reinforce or use thicker pipe insulation (20mm or thicker). Otherwise condensation may form on the surface of the insulating material.

**** Where additional refrigerant is required, be sure to use only R32. The use of other types of refrigerant could cause product malfunction, refrigerant leak or explosion. ****

⚠ CAUTION

- Avoid introduction of air or other contaminants into the refrigerant piping. If any air, moisture or other contaminant remains in the refrigerant piping after the completion of installation, the system will malfunction.
- Carry out sufficient ventilation if the refrigerant gas leaks during installation work.
- Use only R32 rated copper seamless tube manufactured to AS/NZS1571.

- Be sure to use both a spanner and torque wrench together, as shown in the diagram, when connecting or disconnecting pipes to/from the unit. (Fig. 7)
- Use only a flaring block suitable for R32 tube. Refer to Table 2 for the specifications of the flare work.
- When connecting the flare nut, coat ether or ester oil only to flare inner surface, and initially tighten by hand 3 to 4 turns before tightening firmly. (Fig. 8)
- Use the flare nuts included with the indoor unit.
- Refer to Table 2 to determine the proper tightening torque.

⚠ CAUTION

- Over tightening the flare nuts may cause the flare nuts or copper tube to fail and allow the refrigerant to leak.

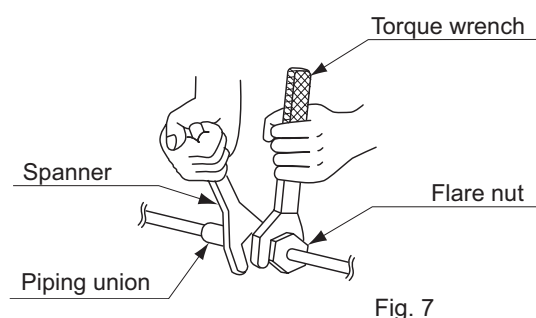


Fig. 7

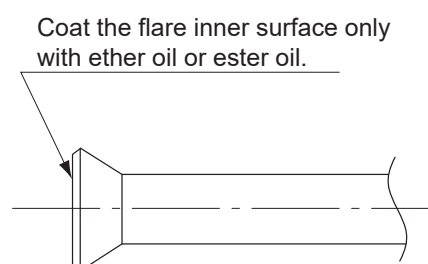


Fig. 8

• Caution to be taken when brazing refrigerant piping. (Where applicable).

Do not use flux when brazing refrigerant piping. Use the phosphor copper brazing filler metal (BCuP) which does not require flux. (Flux has an extremely harmful influence refrigerant piping systems. For instance, if chlorine based flux is used, it will cause pipe corrosion, particularly if the flux contains fluorine, it will damage the refrigerant oil).

During any brazing of the refrigerant piping, you must continually blow dry nitrogen gas through the piping to expel air from the piping. (Fig. 9) If you braise the piping without doing so a large amount of oxide film develops inside the piping and could cause system failure.

Ensure that you use a pressure reducing valve on the dry nitrogen bottle to prevent pinholes caused by excessively high pressure. (Set the pressure of nitrogen gas to about 0.2 kg/cm² [19.6 kPa])

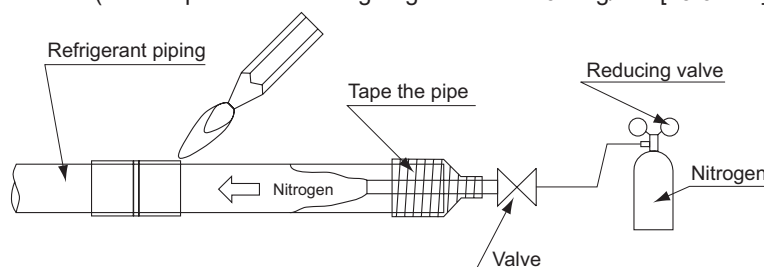


Fig. 9

Table 2

Pipe size	Tightening torque	Flare dimensions A (mm)	Flare
6.4 (1/4")	14.2 - 17.2 N·m (144 - 175 kgf·cm)	8.6 - 9.0	
9.5 (3/8")	32.7 - 39.9 N·m (333 - 407 kgf·cm)	12.6 - 13.0	
12.7 (1/2")	49.5 - 60.3 N·m (504 - 614 kgf·cm)	15.8 - 16.2	
15.9 (5/8")	61.8 - 75.4 N·m (630 - 770 kgf·cm)	19.3 - 19.7	

Not recommended but in case of an emergency

You must use a torque wrench but if you are obliged to install the unit without a torque wrench, you may follow the installation method below.

As you keep on tightening the flare nut with a spanner there is a point where the tightening torque suddenly increases. From that position, further tighten the flare nut using the angle shown below in Table 3:

After the work is finished, make sure to check that there is no gas leak.

Table 3

Pipe size	Further tightening angle	Recommended arm length of tool
6.4 (1/4")	60 to 90 degrees	Approx. 150mm
9.5 (3/8")	60 to 90 degrees	Approx. 200mm
12.7 (1/2")	30 to 60 degrees	Approx. 250mm
15.9 (5/8")	30 to 60 degrees	Approx. 300mm

Make absolutely sure to properly insulate the connecting pipe work and connections using field supplied insulation after checking for gas leakage.

⚠ CAUTION

Be sure to insulate local piping all the way into the pipe connections inside the machine. Exposed piping may cause condensation or burns on contact.

DRAIN PIPING WORK

⚠ WARNING Install the drain pipe as shown below and take measures against condensation, improperly installed piping may result in water leaks and eventually wet furniture and belongings.

(1) Installing

- Route the pipes so as to ensure that drainage can occur with no problems. (Fig. 10)
- Employ piping with either the same diameter or with the diameter larger (excluding the raised section) than that of the connecting pipe (PVC pipe, nominal diameter 25mm, outside diameter 32mm).
- Keep the drain pipe short and sloping downward at a gradient of at least 1/100 to prevent air pockets from forming.
- To prevent the drain pipe from sagging, secure the pipe every 1 to 1.5m.
- Use the provided drain hose ② and clamp ①. (Fig. 11)
Tighten the clamp firmly.
- Insulate the metal clamp with a field supplied sealing pad (Fig. 11)
- Insulate the drain hose inside the building.
- As a safety measure, please use a safety drain tray (field supplied) and drain separately.
- Insulate drain pipe using piping thermal insulation to prevent condensate dripping during humid conditions.

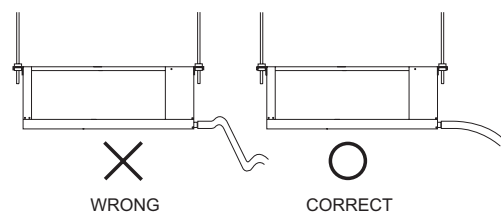


Fig. 10

Note

- To ensure no excessive pressure is applied to the included drain hose ②, do not bend or twist when installing. (This may cause leakage).

⚠ CAUTION

Drain piping connections.

Do not connect the drain piping directly to sewage pipes. Drainage must conform to local plumbing regulations. The vapour in the sewage might enter the indoor unit through the drain pipes and corrode the heat exchanger or escape into the room.

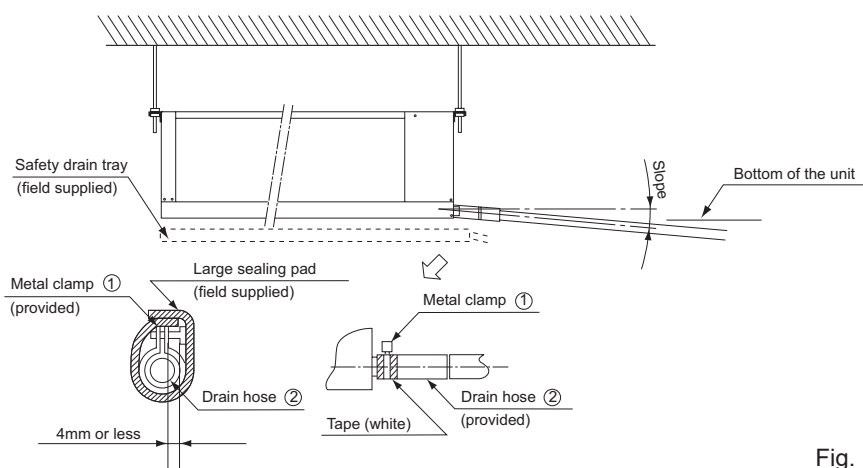


Fig. 11

NOTE:

- If installing central drain piping, install according to Fig. 12.

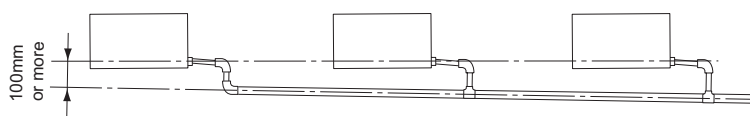


Fig. 12

- (2) After the piping work is finished, check that the drainage flows smoothly.

- Pour approximately 1 Litre of water gradually into the drain tray and check drainage flow. (Fig. 13)

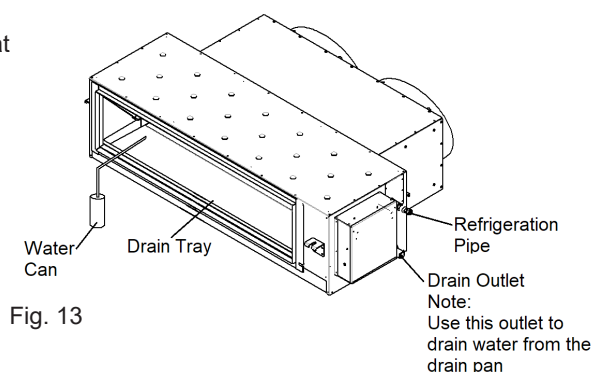


Fig. 13

WHEN ELECTRIC WIRING WORK IS FINISHED

- Check drainage flow during COOLING OPERATION, explained under the section entitled TEST OPERATION.

ELECTRICAL WIRING WORK

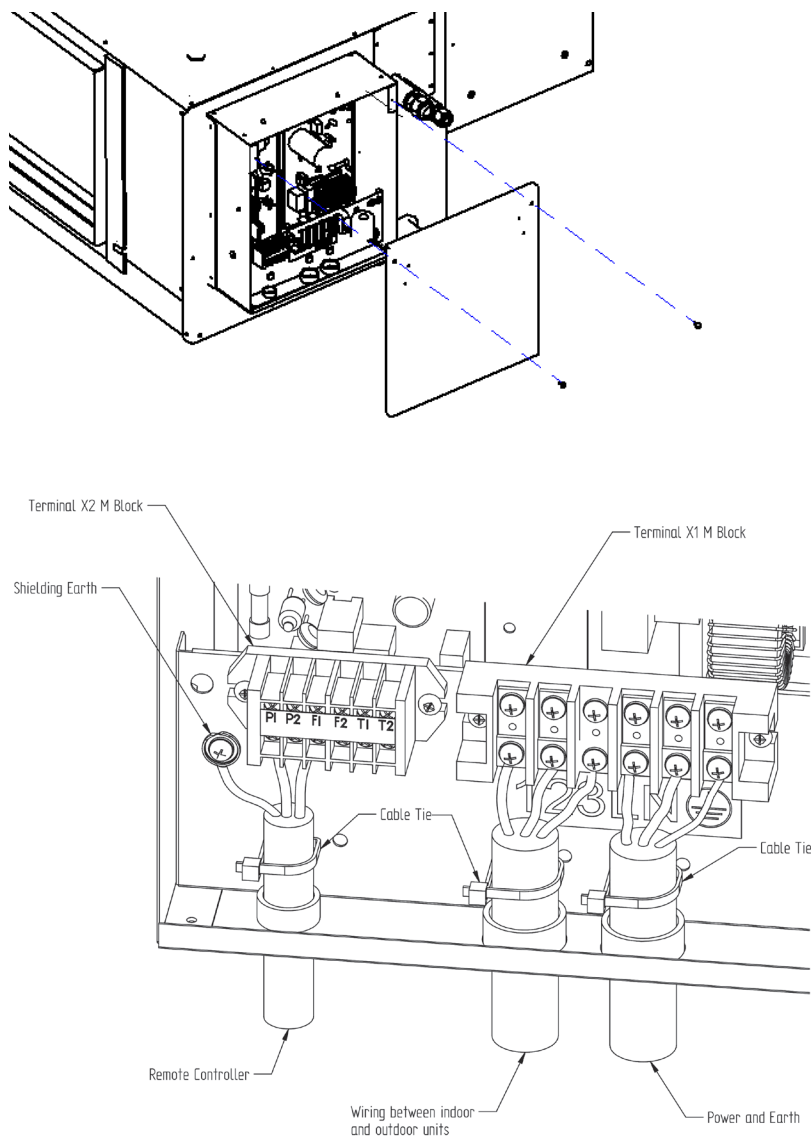
- All field supplied parts, materials and electric works must conform to local codes. (ie. AS/NZS 3000)
- Use copper wire only.
- For electric wiring work, refer to the "WIRING DIAGRAM" label attached to the electrical box cover.
- For remote controller wiring details, refer to page 13 of this manual for BRC1E63 or the installation manual supplied with the chosen remote controller.
- All wiring must be performed by an authorised electrician in accordance with local codes. (ie. AS/NZS 3000)
- A circuit breaker capable of shutting down power supply to the entire system must be installed.
- Select and install the power supply wiring and transmission wiring in accordance with local standards and regulations. The maximum current of the outdoor and indoor unit are shown on each name plate.
- Ensure that the air conditioner is effectively earthed.

- NOTE:**
1. Shield wire materials may be used for transmission wiring, and must comply with EMI (AS/NZS 3548) or (CISPR 14).
 2. Use double insulation wire for the remote controller (Sheath thickness is 1mm or more) or run wires through a wall or conduit so that the user cannot come in contact with them.
 3. Run indoor to outdoor transmission wiring and indoor power supply wiring through a conduit to protect against external forces. Feed the conduit through the wall together with refrigeration piping.

Method of wiring. - Refer to Fig. 14 for the wiring details.

NOTE: All wires must be wired through bushes at the bottom of the electrical box.

- Remove the electrical box cover as indicated.
- Power and earth wiring.
Connect the power supply wires to terminals (L, N). and earth wire to terminal $\perp$.
Secure the cable to the anchor point below the terminal block. Do not connect the N-line with the earth terminal.
- Wiring between indoor and outdoor units.
Connect the transmission wire between the indoor and outdoor unit using terminals (1, 2, 3).
Secure the cable to the anchor point below the terminal block.
- Remote controller cords.
Connect the cords to the remote controller terminals - (P1, P2 No polarity) and the shield wire (if applicable) to the shielding earth terminal $\perp$.
Secure the cable to the anchor point below the terminal block.
- Replace and secure electrical box cover.



FDYAN50-160AV1

FDYA71-160AV1

Fig. 14

PRECAUTIONS

Precaution for wiring on site

When connecting the indoor unit to a single phase of 3 phase power supply, give consideration to the other electrical equipment connected to the main breaker and select the phase (L1, L2, L3) so that the current (or voltage) unbalance does not occur.

If current unbalance occurs, the phase current of the compressor may become unbalanced and there is a risk of protective device actuating or compressor breakdown.

- Do not clamp remote controller cords together with wires connecting the units. Doing so may cause malfunction.
- Remote controller cords and wires connecting the units should be located at least 50mm from other electric wires. Not following this guideline may result in malfunction due to electrical noise.

Observe the notes mentioned below when wiring to the power supply terminal block. Precautions to be taken for power supply wiring.

(Use a round crimp-style terminal for connection to the power supply terminal block. (Fig. 15) In case it cannot be used due to unavoidable reasons, be sure to observe the following instructions.)

- Do not connect wires of different gauge to the same power supply terminal. (Fig. 16)
(Looseness in the connection may cause overheating.)
- When connecting wires of the same gauge, connect them according to the figure below.
- Make certain that prescribed wires are used, carry out complete connections, and secure the wires so that outside forces are not applied to the terminals.

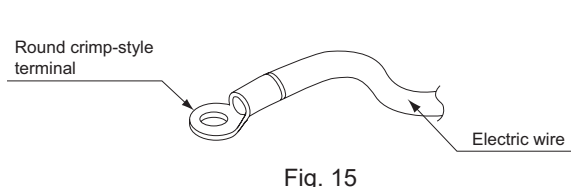


Fig. 15

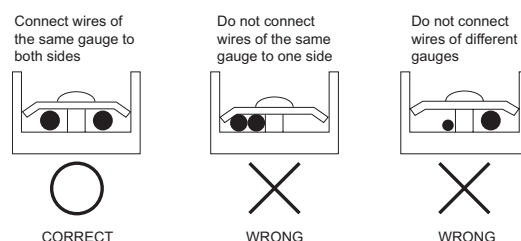


Fig. 16

⚠ CAUTION

- After connecting the wires, be sure that no pressure is applied to the wire connections by using cable ties (field supplied) and securing them to the anchor points. Also, when wiring, make sure the cover of the terminal box fits snugly by arranging the wires neatly and attaching the terminal box cover firmly. When attaching the terminal box cover make sure no wires get caught under the edges. Ensure that all the cables pass under the offset section of the electrical box cover.
- Make sure the remote control cord, the wiring between the units, and other electrical wiring do not pass through the same locations outside the unit, separating them by at least 50mm, otherwise electrical noise (external static) could cause incorrect operation of the unit or damage.

WIRING EXAMPLE for BRC1E63 only

(For the wiring of outdoor units, refer to the installation manual attached to the outdoor unit.)
Confirm the system type.

- Single phase pair type: 1 remote controller controls 1 indoor unit (standard system). (Fig. 17)
- 2 remote controller control: 2 remote controller controls 1 indoor unit. (Fig. 18)
- Group control: 1 remote controller controls up to 16 indoor units (All indoor units operate according to the remote controller). (Fig. 19)

Single phase pair type

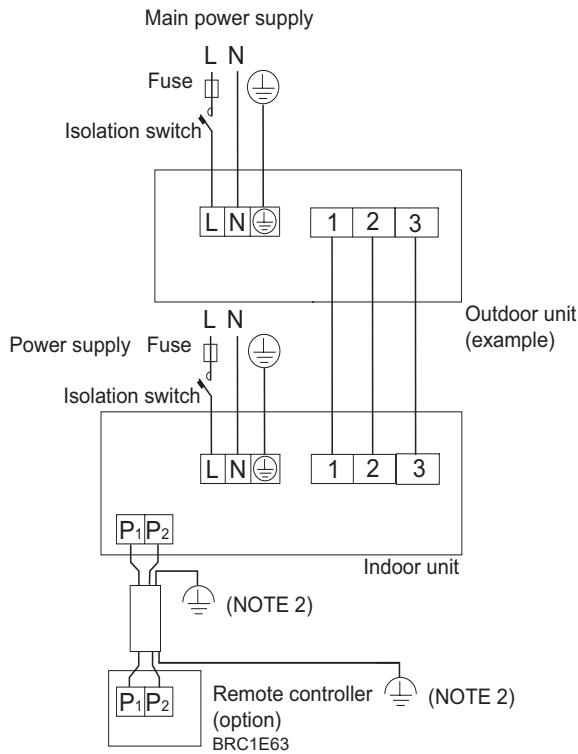


Fig. 17

Control by 2 remote controllers

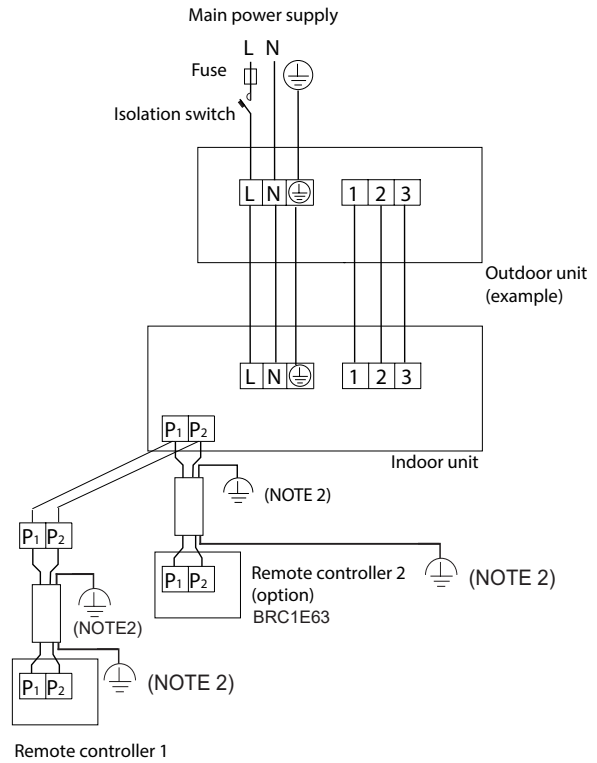


Fig. 18

Group Control (Only Applies to BRC1E63)

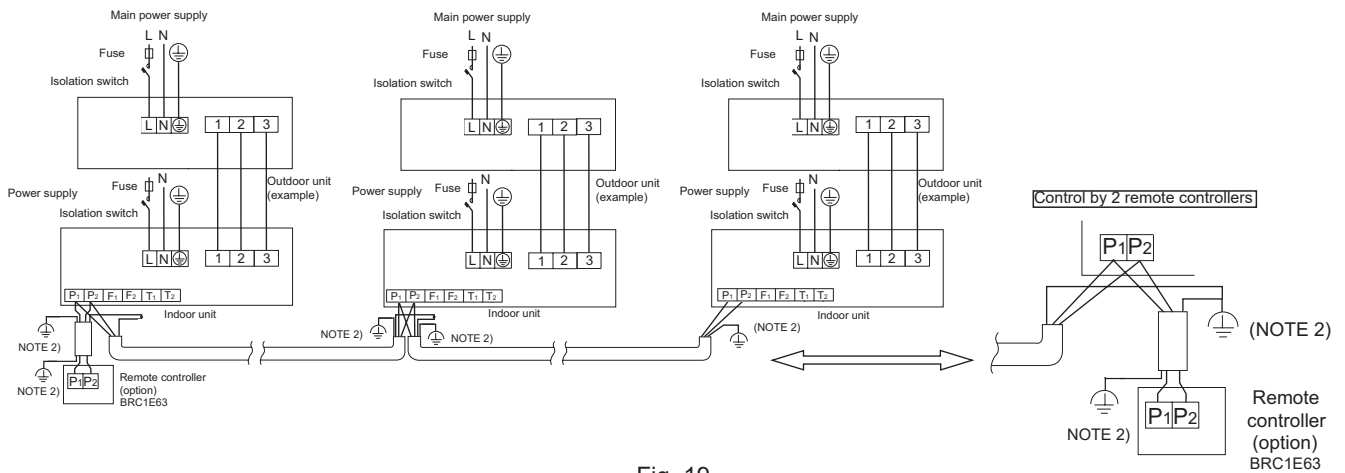


Fig. 19

- NOTE: 1. All transmission wiring except for the remote controller wires must match the terminal symbol.
2. If using shielded wire for transmission wiring ground the shield of the shield wire to  at the grounding screw of the remote controller cord grounding terminal inside the control box.
3. For group control remote controller choose the remote controller that suits the indoor unit which has the most functions (as indicated by the swing flap).
- Only applies when other types of indoor units are used within the group eg. Cassette type.
4. When "Group Control" and "Control by 2 remote controllers" are used, connect the remote controller with the indoor unit which is at the end. (Fig. 19)
(Do not connect three wires or more to the remote control terminals.)

Field Settings

How to Change the Field Settings with the Wired Remote Controller

The following functions of indoor unit can be changed from the remote controller. At the item of installation or after service inspection / repair, make the local setting in accordance with the following description.

Wrong setting may cause malfunction.

- Setting indoor fan target speed
- Setting air filter sign
- Setting temperature sensor location
- Indoor fan setting during heating thermo-off
- Indoor fan setting during cooling thermo-off
- Indoor fan setting during heating/cooling
- Setting for T1, T2 input
- Setting for thermostat differential
- Setting for thermostat sensor in remote
- Setting for permission level
- Setting for home leave function

(When optional accessory is mounted on the indoor unit, setting for the indoor unit may be required to change. Refer to information in the option handbook.)

Wired Remote Controller <BRC1E63>

- 1 Press and hold the Cancel button for 4 seconds or longer. Service Settings menu is displayed.

- 2 Select **Field settings** in the Service Settings menu, and press Menu/Enter button. Field settings screen is displayed.

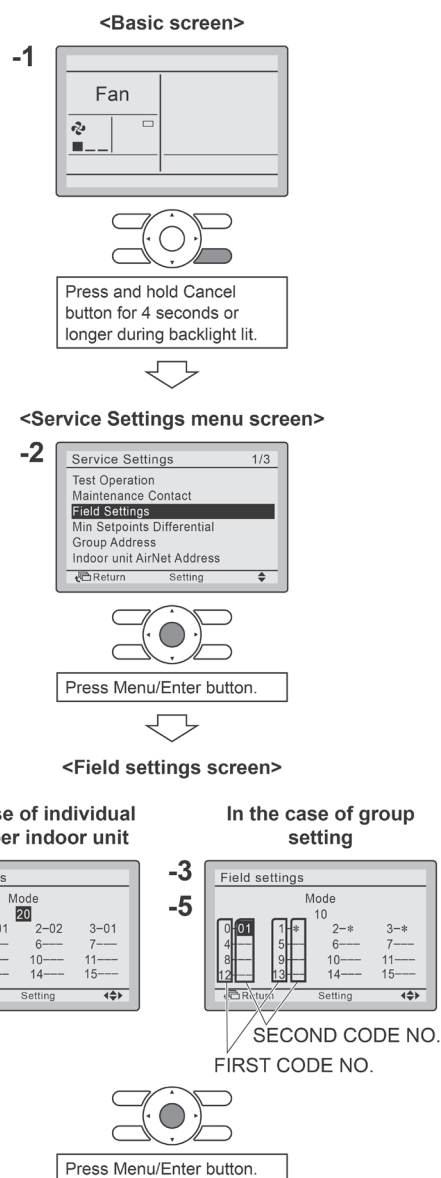
- 3 Highlight the Mode No., and select desired "Mode No." by using "▼▲" (Down/Up) button. For the commonly-used field setting item, refer to page 14 and 15.

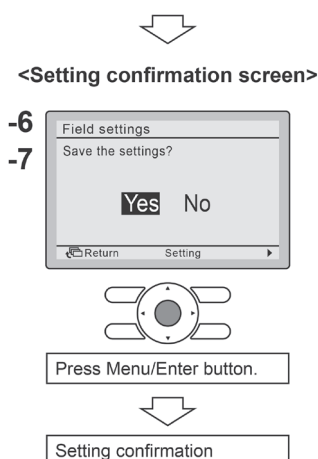
- 4 In the case of setting per indoor unit during group control (When Mode No. such as **20**, **21**, **22**, **23**, **25** are selected), "Indoor unit No." to be set by using "▼▲" (Down/Up) button. (In case of group setting, this operation is not needed.)

In case of individual setting per indoor unit, current settings are displayed. And, SECOND CODE NO. " - " means no function.

- 5 Highlight SECOND CODE NO. of the FIRST CODE NO. to be changed, and select desired "SECOND CODE NO." by using "▼▲" (Down/Up) button. Multiple identical mode number settings are available.

In case of group setting, all of SECOND CODE NO. which may be set are displayed as " * ". " * " is changed to SECOND CODE NO. to be set. And, SECOND CODE NO. " - " means no function.





- 6** Press Menu/Enter button. Setting confirmation screen is displayed.
- 7** Select **Yes** and press Menu/Enter button. Setting details are determined
- 8** In the case of multiple setting changes, repeat “(3)” to “(7)”.
- 9** After all setting changes are completed, press Cancel button twice.
- 10** Backlight goes out, and “Connection under check Please wait for a moment” is displayed for initialization. After the initialization, the basic screen returns.

⚠ CAUTION

- When an optional accessory is installed on the indoor unit, settings of the indoor unit may be changed. See the manual of the optional accessory.
- For field setting details of the outdoor unit, see installation manual attached to the outdoor unit.

FAN MANUAL SETTING

Daikin FDYA-AV1 Series Ducted Fan Coil units employ a high efficiency DC Fan Motor operating in combination with a Daikin proprietary microprocessor based motor management system. The DC Fan Motor contains multiple motion sensor devices that monitor the motor’s rotation and speed. The output from these devices is constantly compared against target data in the motor management system which subsequently makes adjustments to the motor input waveform to maintain the target motor speed.

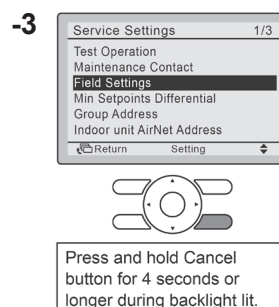
To match the airflow requirements to the individual installation, the target fan motor speed can be commissioned via the remote control to one of fifteen individual settings as described below. The default factory setting is No. “10” so if you experience excessive airflow during post installation checks please follow the field setting instructions to set a lower airflow.

Please note that any change to the fan motor speed via the method described below is applied across all 3 user selectable speeds (ie High, Medium, Low).

Setting indoor fan target speed		(Second Code No. is factory set to “10”)	
Setting	Mode No.	First Code No.	Second Code No.
Maximum	13 (23)	6	15
Between maximum and minimum			02-03.....14
Minimum			1

HOW TO SET FAN AUTOMATIC ADJUSTMENT

- 1** Set the controller to “Fan” mode.
- 2** Ensure that the unit is Switched off/ Standby mode.
- 3** Press and hold the Cancel button for 4 seconds or longer until **Field settings** appears.



-4 Press “▼▲” (Down/Up) button until Mode **21** appears.

-5 Press “◀▶” (Right/Left) button to go to **7-01**.

-6 Change settings to **7-03** by pressing “▼▲” (Down/Up) button.

-7 Press Menu/enter to confirm then select “Yes” to accept changes.

-8 Press Cancel button twice to return to main display.
Note: The controller will initialize for the changes made.

-9 Turn on unit to start Automatic Airflow Adjustment.

- The LED on the controller should turn on and the indoor unit will run on different rpm for a period of 2-10 minutes.
- When the Indoor unit stops and LED on the controller turns off, it means that the Automatic Airflow Adjustment is finished.

Note: Meanwhile do not press any button.

-10 To confirm that the Automatic Airflow Adjustment is successful do steps 3 to 4.

7-02 should appear, if not set the Automatic Airflow Adjustment again.

Note: 1. If the indoor unit does not stop in the period of 10 minutes or 7-02 does not appear, it means that Automatic Airflow Adjustment has failed and need to be set again.

2. If setting the Automatic Airflow Adjustment fails again on the second try, the rpm need to be setup manually.

7-01	Turn OFF Auto Airflow Adjustment
7-02	Airflow Adjustment complete
7-03	Airflow Adjustment Initialization

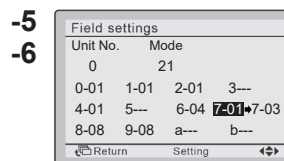
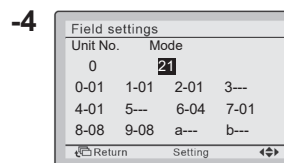
-11 After confirming that the Automatic Airflow Adjustment is successful, normal operation can now commence.

Possible reasons why Automatic Airflow Adjustment is failing.

1. External static is too low.
2. Duct may not be stable, this happens when using flexible duct and not properly fixed in place.



<Field settings screen>



OTHER SETTINGS (Applicable to all models)

Setting air filter sign			
Setting	Mode No.	First Code No.	Second Code No.
Filter sign indication [ON]	10 (20)	3	01
Filter sign indication [OFF]			02

Setting temperature sensor location			
Settings	Mode No.	First Code No.	Second Code No.
Alternate between indoor unit and remote controller sensor	10(20)	2	01
Indoor unit sensor only			02
Remote controller sensor only			03

Indoor fan settings during heating thermo off			
Settings	Mode No.	First Code No.	Second Code No.
LL	12(22)	3	01
Set speed			02
OFF			03

Indoor fan settings during cooling thermo off			
Settings	Mode No.	First Code No.	Second Code No.
LL Fan Speed	12(22)	6	01
Set Fan Speed			02
Fan OFF			03

Indoor fan settings during cooling/heating thermo off			
Settings	Mode No.	First Code No.	Second Code No.
Normal	11(21)	2	01
Fan OFF			02

Setting for T1, T2 input			
Setting	Mode No.	First Code No.	Second Code No.
Forced OFF	12 (22)	1	01
ON/OFF operation			02

Setting for thermostat sensor in remote controller			
Setting	Mode No.	First Code No.	Second Code No.
Do Not Use	1c	1	01
Use			02

Setting for permission level			
Setting	Mode No.	First Code No.	Second Code No.
Level 2	1c	3	01
Level 3			02

= Factory Setting

Setback function	Mode No.	First Code No.	Second Code No.
Setting			
Not Used	1e	2	01
Heating Only			02
Cooling Only			03
Cooling and Heating			04

CONTROL BY 2 REMOTE CONTROLLERS

- Check for completion of indoor/outdoor units wiring.
- Check for closing of EL. COMPO. BOX cover of indoor and outdoor units before power-on.

- 1** Following is displayed after turn on the power.
 “Checking the connection.
 Please stand by.”

During above display, backlight does not light by button operation.

When 1 indoor unit is controlled by 2 remote controllers:

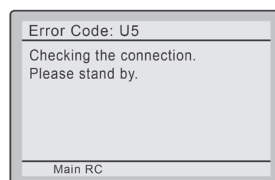
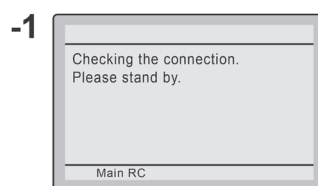
Be sure to set sub remote controller during above display. Press and hold for 4 seconds or longer the Mode Selector button of the remote controller to be set. When the display is changed from main remote controller to sub remote controller, the setting is completed.

- 2** Basic screen is displayed.

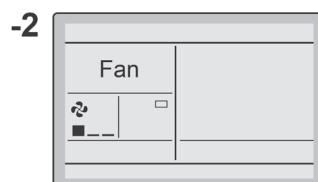
CAUTION

If sub remote controller is not set at turn on the power in the case of 1 indoor unit controlled by 2 remote controllers, "Error code: U5" is displayed in the connection checking screen. Select the sub remote controller by pressing the Mode Selector button of either one of the remote controllers for 4 seconds or longer. If the Basic screen is not displayed more than 2 minutes after "Sub RC" display, turn off the power and check the wiring.

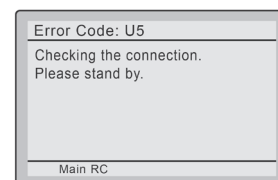
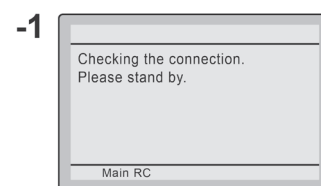
<Main remote controller>



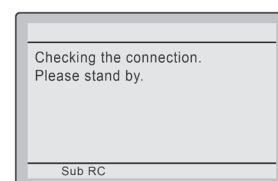
<Basic screen>



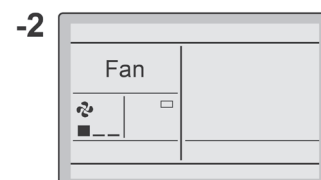
<Sub remote controller>



Press and hold for 4 seconds or longer the Mode Selector button of sub remote controller side.



<Basic screen>



Test Operation

- Please refer to the instruction manual for remote controller (BRC1E63)

SEPARATING THE INDOOR UNIT FOR ALL MODELS

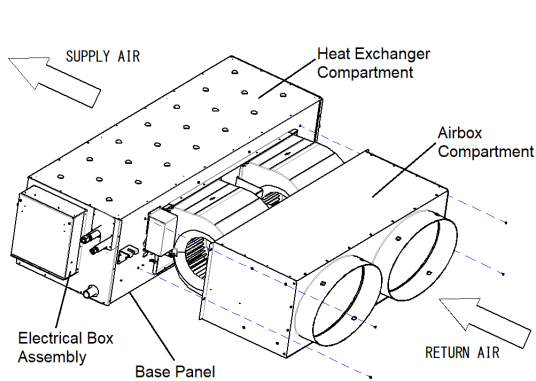


Fig. 20

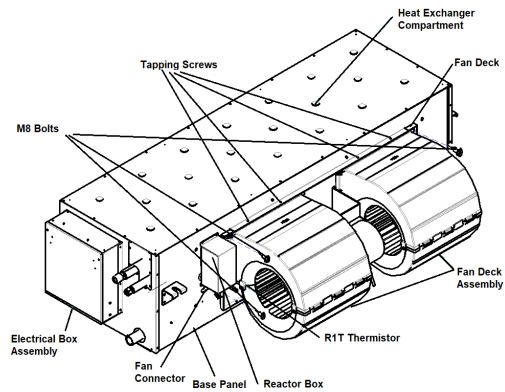


Fig. 21

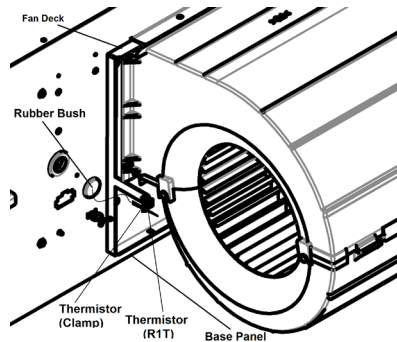


Fig. 22

⚠ WARNING: Power should be disconnected and tagged off when performing the following procedure when the unit is already installed.

ATTENTION: Please separate the indoor unit only when necessary. Please disassemble and reassemble on a flat surface.

Separate the indoor unit in the following order.

- 1) Carefully remove the Airbox compartment by removing the (4) M6 bolts (Fig.20).
- 2) Carefully remove the screw on the thermistor clamp and remove the (1) thermistor cables (R1T). (Fig 22).
- 3) Wrap the thermistor cable and place it neatly into the rubber bush next to the fan deck. (Fig 22).
- 4) Carefully disconnect the fan motor wiring from the base panel located underneath the reactor box. (Fig 21).
- 5) Remove the (4) tapping screws located above the fan deck flange. (Fig 21).
- 6) Remove (4) M8 bolts securing the fan deck to the base panel of the heat exchanger compartment (Fig 21).
- 7) Carefully separate the fan deck assembly from the base panel, taking care not to damage the fan motor wiring. (Fig 21).
- 8) Carefully detach the fan deck. (Fig 21).

• Reassemble the fan deck and heat exchanger compartments in the following order:

- 9) Carefully place the fan deck back in position to the base panel. (Fig 21).
- 10) Secure the fan deck onto the heat exchanger compartment with the (4) M8 bolts and (4) tapping screws for the top flange of the fan deck. (Fig 21).
- 11) Place the (1) thermistor cable (R1T) back to the side of the fan deck and secure it with the screw and themistor clamp. (Please make sure that tip of the thermistor is position horizontally opposite to the fan casing. (Fig 22).
- 12) Reconnect the fan motor connector back to the connector of the base panel.
- 13) Carefully place the fan deck compartment with (4) M6 bolts, taking care not to damage the fan motor wiring. (Fig 20).

ERROR CODE LIST

Contents Code Section Code		0	1	2	3	4	5	6	7	8	9	A	H	C	J	E	F
INDOOR	A	External protection device activated	Malfunction of EEPROM on indoor unit PCB	Fan interlock (manual)	Abnormal drain water level or plug missing	Abnormal heat exchanger (1) temp.	Abnormal heat exchanger (2) temp.	Malfunction of fan motor	Air direction adjust motor lock	Malfunction of power supply or AC input overcurrent	Faulty electronic expansion valve	Heater overheating	Faulty or dirty air cleaner	No load up	Malfunction of capacity setting (indoor unit PCB)	Insufficient water supply	Drain level above limit
	C	Malfunction of indoor fan driver PCB, Fan driver not defined	Failure of transmission between indoor unit PCB and fan PCB		Faulty drain water level sensor system	Faulty heat exchanger temp. sensor system (R2T)	Faulty gas pipe temp. sensor system (R3T)	Malfunction of fan motor sensor or fan control driver or PCB over temperature.	Air direction motor lock, faulty sensor system	Faulty AC input current sensor	Faulty air suction temp. sensor system (R1T)	Faulty air suction temp. sensor system	Faulty or dirty sensor	Faulty humidity sensor system	Faulty temp. sensor (remote controller)	Faulty radiation sensor system	Faulty high pressure switch
	E	Activation of safety device (general)	Faulty PCB outdoor		Abnormal high pressure (HPS)	Abnormal low pressure (LPS)	Compressor motor overload	Compressor motor over-current lock	Out door fan motor overload over-current lock	AC input over-current	Faulty electronic expansion valve		Pump over-current lock	Abnormal water temp.	Activation of field added safety device	Faulty drain water level sensor system	Faulty heat storage unit
	H	Faulty sensor system (general)	Faulty air temp. sensor system	Faulty power supply sensor system	Faulty high pressure switch	Faulty low pressure switch	Compressor motor faulty overload sensor system	Compressor motor over-current lock faulty sensor system	Abnormal outdoor fan motor signal	Faulty AC input power sensor system	Faulty outdoor air temp. sensor system	Faulty discharge air temp. sensor system	Pump over-current lock faulty sensor system	Faulty hot water sensor system		Faulty drain water level sensor system	Heat storage unit alarm
OUTDOOR	F	Activation of common safety device for system No. 1 & No. 2	Activation of safety device for system No. 1	Activation of safety device for system No. 2	Faulty discharge pipe temp.			Refrigerant overcharge				Abnormal discharge pressure	Abnormal high oil temp.	Abnormal suction pressure		Abnormal oil pressure	Abnormal surface oil level
	J	Faulty refrigerant temp. sensor system	Faulty pressure sensor system (general)	Faulty current sensor system	Faulty discharge pipe temp. sensor system	Faulty low pressure equivalent saturation temp. sensor system	Faulty suction pipe temp. sensor system	Faulty heat exchanger temp. sensor system (R4T)	Faulty receiver outlet liquid pipe sensor (R7T)	Faulty liquid pipe temp. sensor system	Faulty subcooler heat exchanger gas pipe sensor (R5T)	Faulty discharge pressure sensor system	Faulty oil temp. sensor system	Faulty suction pressure sensor system		Faulty oil pressure sensor system	Faulty surface oil level sensor system
	L	Faulty INV system	Faulty PCB		Abnormal temp. rise inside control box	Abnormal temp. rise of radiation fin (power transistor)	DC output over-current instantaneous	AC output over-current instantaneous	Air conditioner (multi) general input over-current	Inverter current abnormal	Inverter start-up error	Faulty power transistor		Faulty transmission between INV and control PCB			
	P	Gas shortage (ice thermal storage equipment)	High voltage of capacitor in inverter circuit or power imbalance		Faulty temp. rise sensor inside control box	Faulty radiation fin (power transistor) temp. rise sensor system	Faulty DC current sensor system	Faulty AC or DC output sensor system	Air conditioner (multi) general input over-current						Faulty capacity setting		
SYSTEM	U	Short of gas	Reversed phase connection	Faulty power supply voltage	Faulty transmission (general) or check operation is not complete	Failure of transmission between indoor and outdoor units	Faulty transmission between indoor unit and remote controller	Faulty transmission between indoor units (main & sub)	Faulty transmission between outdoor units	Faulty transmission to/from remote controllers (main & sub)	Faulty transmission between indoor & outdoor in the same system	Improper combination of indoor & outdoor, indoor & remote control	Indoor/outdoor unit address not entered	Faulty central control address setting (Address duplications)	Faulty transmission to/from peripheral equipment (air control equipment, etc)	Faulty transmission between indoor unit and central controller	Wiring and piping mismatch

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